

SURYAN RAJ

suryanraj056@gmail.com | +91 6205931828 | Chirkunda, Dhanbad
GitHub: <https://github.com/SuryanRaj> | LinkedIn: <https://linkedin.com/in/suryan-raj-094571388>

EDUCATION

Asansol Engineering College

Expected Graduation: 2030

Bachelor of Technology

ISC Board (12th Grade)

Score: 91%

ICSE Board (10th Grade)

Score: 90%

TECHNICAL SKILLS

Languages: Native C++, Java, C

Systems Architecture: Low-Latency Design, Bare-Metal Memory Management, Pointers/References, State Machines

Data Structures: Standard Template Library (STL), Red-Black Trees (std::map), FIFO Queues, Big-O Optimization

Quantitative Finance: Market Microstructure, Limit Order Books, Price-Time Priority, Geometric Brownian Motion, Box-Muller Transforms, Stochastic Calculus

ENGINEERING PROJECTS

High-Frequency Trading Limit Order Book (LOB) Engine

Native C++, STL

<https://github.com/SuryanRaj/Limit-Order-Book-LOB>

- Engineered a low-latency matching engine simulating the core order-routing architecture of modern financial exchanges.
- Architected a dual-sided book leveraging Red-Black Trees (std::map) for $O(\log N)$ insertions, alongside FIFO Queues (std::queue) to enforce Price-Time Priority.
- Implemented memory-safe execution loops utilizing direct memory references to process full and partial order fills without memory leaks.
- Stress-tested the state machine using a Mersenne Twister PRNG to process high-volume, Bell-Curve distributed market orders.

Monte Carlo Stochastic Stock Simulator

C++, Quantitative Math

https://github.com/SuryanRaj/monte_carlo-simulator

- Forecasted equity prices by modeling market volatility via stochastic calculus, Geometric Brownian Motion (GBM), and Ito's Lemma.
- Optimized algorithmic loops to calculate 2.5+ million daily price points, generating 10,000 alternate market trajectories over a 252-day period.
- Extracted hardware-level entropy (std::random_device) passed through a Box-Muller transform to simulate real-world market distributions.

Algorithmic SMA Financial Backtester

C++, I/O Streams

<https://github.com/SuryanRaj/cpp-sma-backtester>

- Developed a command-line backtesting engine to evaluate quantitative trading strategies using custom moving window mathematics.

- Engineered a memory-efficient CSV parser using standard C++ streams to ingest and process chronological market datasets.
- Simulated algorithmic portfolio management, handling virtual asset accumulation, dynamic cash deductions, and exact ROI calculations.